

Joe Lopez

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WORK EXPERIENCE

Microsoft Corporation

Redmond, WA

FIRMWARE DEVELOPMENT ENGINEER

SEP 2021 – JUL 2023

- Developed UEFI firmware for PCIe hardware initialization on company's first purpose-built ARM server platform, Cobalt 100. Ported open source libraries and contributed fixes upstream.
- Mitigated boot issues during chip and platform bringup by diagnosing ACPI issues using Windows Debugger (WinDbg) and ACPI tools, releasing a firmware patch in a timely manner to unblock others.
- Led the integration of firmware unit test suite into platform codebase, consuming open source framework and modifying it for our usecase. Worked with internal open-source owners to upstream enhancements for consumption by all first-party and third-party UEFI teams.
- Integrated pipeline for unit test CI/CD into platform codebase, using code coverage tools to compare coverage statistics against old version.
- Enhanced platform Reliability, Availability, and Serviceability (RAS) by constructing drivers for PCIe error reporting and handling, including workarounds for various silicon errata.

SOFTWARE DEVELOPMENT ENGINEER 2

JUL 2023 – MAY 2025

- Examined codebase of first-generation product to plan and implement second-gen firmware based on new hardware layout. Developed design plan and distributed work amongst vendor team and oversaw firmware delivery timeline.
- Leveraged internally developed tool and agentic AI to generate firmware unit tests for previously uncovered lines of code using techniques such as RAG and multi-model approaches to achieve higher efficiency and scale to larger code files.
- Induced cross-team collaboration by facilitating work forum for PCIe/CXL hardware and software stakeholders for design reviews, bug investigations, and tech talks.

Robert Bosch Tool Corporation (Dremel)

Mount Prospect, IL

ELECTRICAL ENGINEERING INTERN

MAY 2019 – AUG 2019

- Collaborated with Electrical and Mechanical engineers to create prototype tool with USB (UART) interface to decrease reliance on bluetooth chips
- Improved data acquisition capacity by rewriting firmware for custom tool logger
- Assembled and diagnosed issues on prototype tools using oscilloscope, multimeter

EDUCATION

University of Illinois

Urbana-Champaign, IL

BACHELOR'S IN COMPUTER ENGINEERING

AUG 2017 – JUN 2021

- Relevant coursework in Systems Programming, Computer Science, Silicon

LEADERSHIP EXPERIENCE

Illini Formula Electric

DATA ACQUISITION LEAD

AUG 2019 – AUG 2020

- Revised effectiveness of the previous data acquisition system in order to design a system that logs vehicle data with LORA radio, GPS, Wi-Fi, and serial communication.
- Optimized costs and performance by designing multiple custom PCB's with Altium and implemented logging firmware/software between microcontroller and Raspberry Pi.

Theta Tau Professional Engineering Fraternity

PROJECTS COMMITTEE CHAIR

AUG 2019 – JUN 2020

- Managed a committee of 18 members to develop a project for presentation
- Introduced new technologies to the fraternity by hosting a microcontroller workshop

SKILLS

Programming Languages: C, C++, UEFI Framework, Python, Azure CI/CD, System Verilog, x86

Embedded Electronics: ARM, PCIe, STM32, I2C, SPI, UART, CAN

Miscellaneous: Altium Designer, Solidworks, GDB, Git, development in Windows/Linux environments